



SOMAIYA
VIDYAVIHAR UNIVERSITY

Title

Domain :

Category: Software Edition

Team ID: SIH26_19

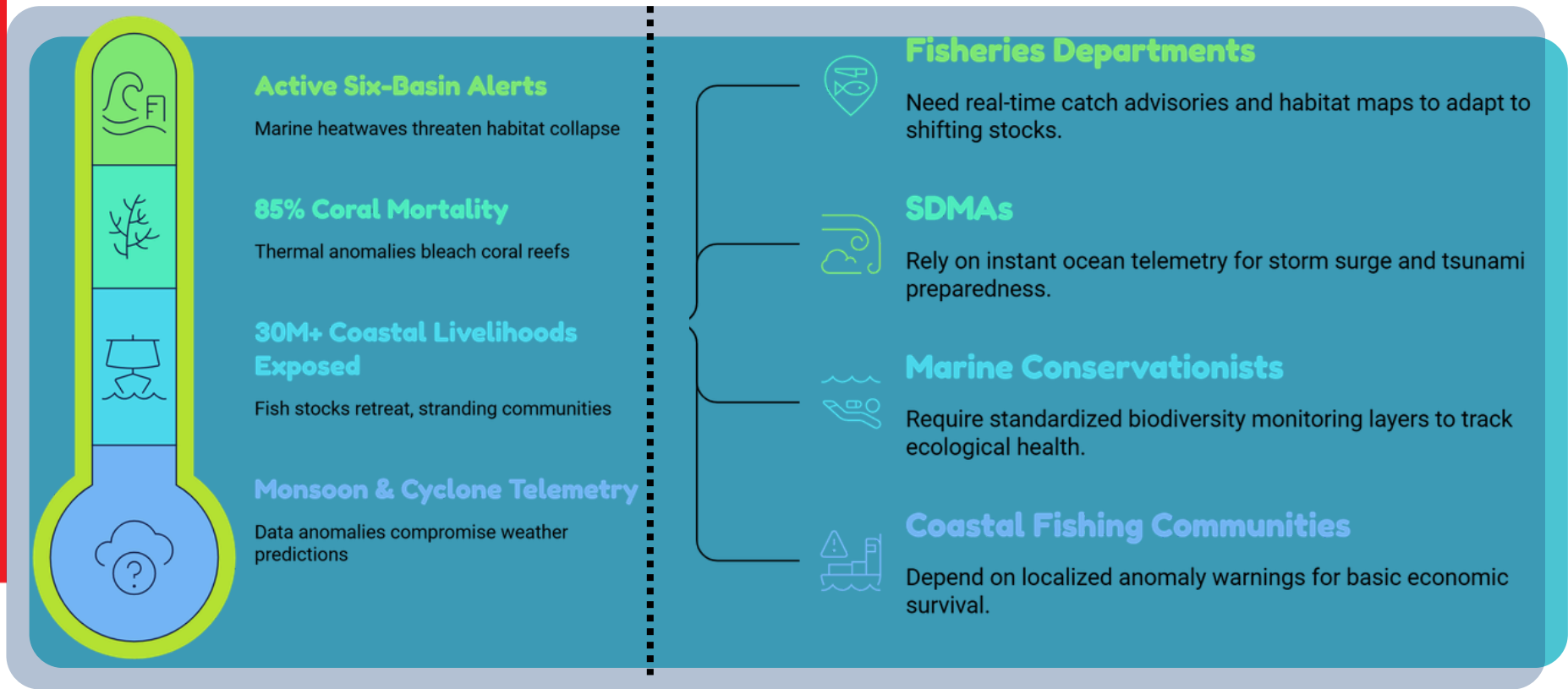
Team Name: Ctrl Alt Defeat

Problem Statement

The lack of a unified, real-time spatial-temporal bridge between INCOIS physical oceanographic observations and CMLRE marine biodiversity records creates a critical data-governance gap that weakens national fisheries policy and disaster preparedness during severe marine anomalies.

Impact of Governance gap in Marine Economies and Ecosystems

Affected Stakeholders and what they really require



Literature Survey & Existing Solutions

Grounding & Prior Work

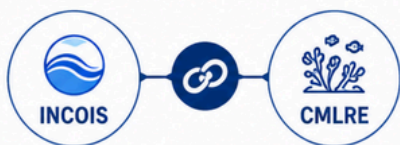
#	Theme	Citation	Our Understanding	Application to VARUNA
1	Deep Learning for Text-to-SQL	Katsogiannis-Meimarakis, G. & Koutrika, G. (2023). A survey on deep learning approaches for text-to-SQL. The VLDB Journal, 32(4), 905–936.	Most production failures come from ambiguous column references and missing schema context at generation time, not from the model's SQL grammar; a validation layer is standard practice.	Every generated query passes through an AST-level sanitizer (SQLGlue-based, SELECT-only, batch-injection blocked, LIMIT ≤500) before execution.
2	Reasoning + Acting in LLMs	Yao, S. et al. (2023). ReAct: Synergizing reasoning and acting in language models. ICLR.	Interleaving explicit reasoning steps with tool-use actions (rather than pure reasoning or pure acting) reduces compounding errors on multi-step tasks by allowing course-correction between steps.	Mirrored in the Synthesizer's verify-before-finalize step, which checks every claim against actual sub-agent outputs rather than committing to an answer in one forward pass.
3	Biodiversity Data Standardization	Wieczorek, J. et al. (2012). Darwin Core: An Evolving Community-Developed Biodiversity Data Standard. PLoS ONE, 7(1), e29715.	A shared vocabulary of terms (scientificName, decimalLatitude, eventDate, etc.) letting independently-collected biodiversity records be aggregated and queried together without a common backend	marine_biodiversity table built directly on Darwin Core terms, so it is already compatible with CMLRE's actual production data format; OBIS/GBIF usable as PoC stand-in with zero format-translation layer.
4	Reviewed Winning Solution	OceanIQ (SIH 2025 winner. PS 25040) — public repository	Solid production engineering: real-time IFREMER/Argo ingestion with DAC detection, PostgreSQL+PostGIS, dual FAISS indices, Gemini-based NL-to-SQL with validator/sanitizer/executor chain, JWT auth, rate limiting.	Gap addressed by VARUNA: single-shot pipeline (no task decomposition), stateless (no session memory), purely reactive (no anomaly detection), single dataset (no biodiversity fusion), no visible frontend, no offline mode.

Explanation of the proposed solution

VARUNA is an agentic AI platform that fuses INCOIS ocean-physical data and CMLRE biodiversity data into one proactive intelligence system.

Unlike a single-shot chatbot, VARUNA's planner agent decomposes compound questions into sub-tasks handled by specialized agents : SQL generation, retrieval, cross-domain entity-resolution, comparison. A dedicated anomaly agent continuously watches incoming ocean data and pushes heatwave/anomaly alerts without waiting to be asked. Every answer is narrative, chart, or map traces back to a source record, and session memory keeps multi-turn conversations coherent.

How VARUNA Solves the Problem



1. Bridges the Silo

Fuses INCOIS ocean-physical data with CMLRE biodiversity data to eliminate the current integration gap.



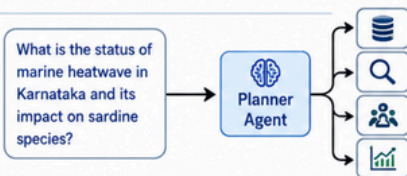
2. Proactive, Not Reactive

An anomaly agent continuously scans incoming data and pushes heatwave alerts before anyone thinks to ask.



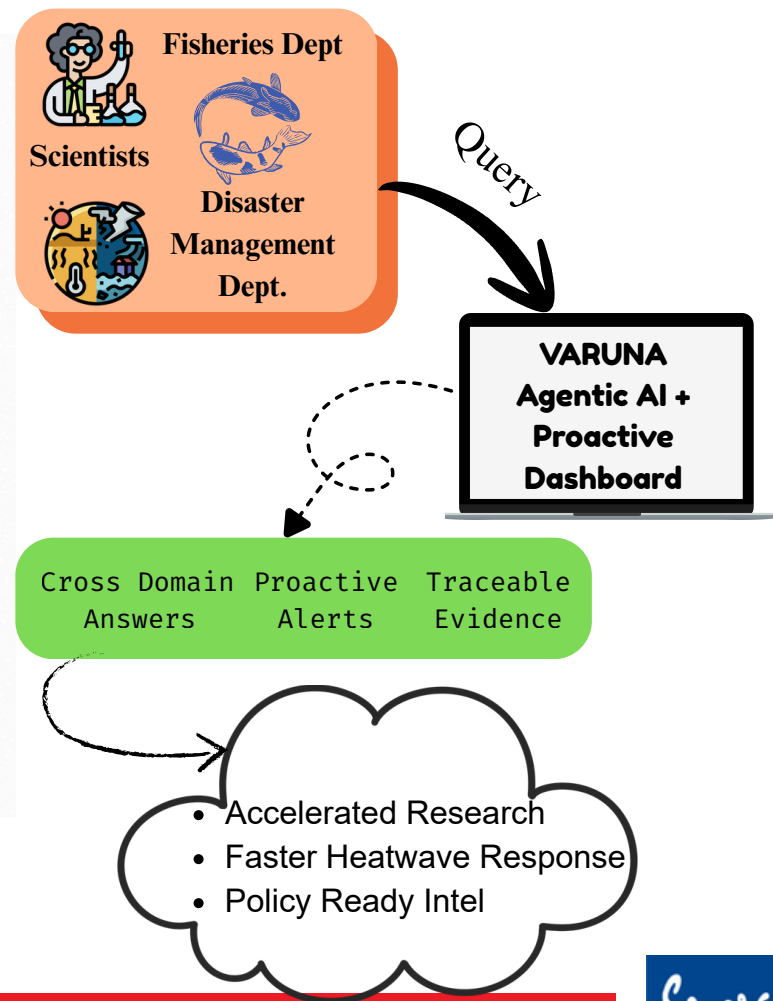
3. Decision-Ready for Non-Experts

Fisheries and disaster-management staff get plain-language source-traceable answers no NetCDF or SQL expertise needed.



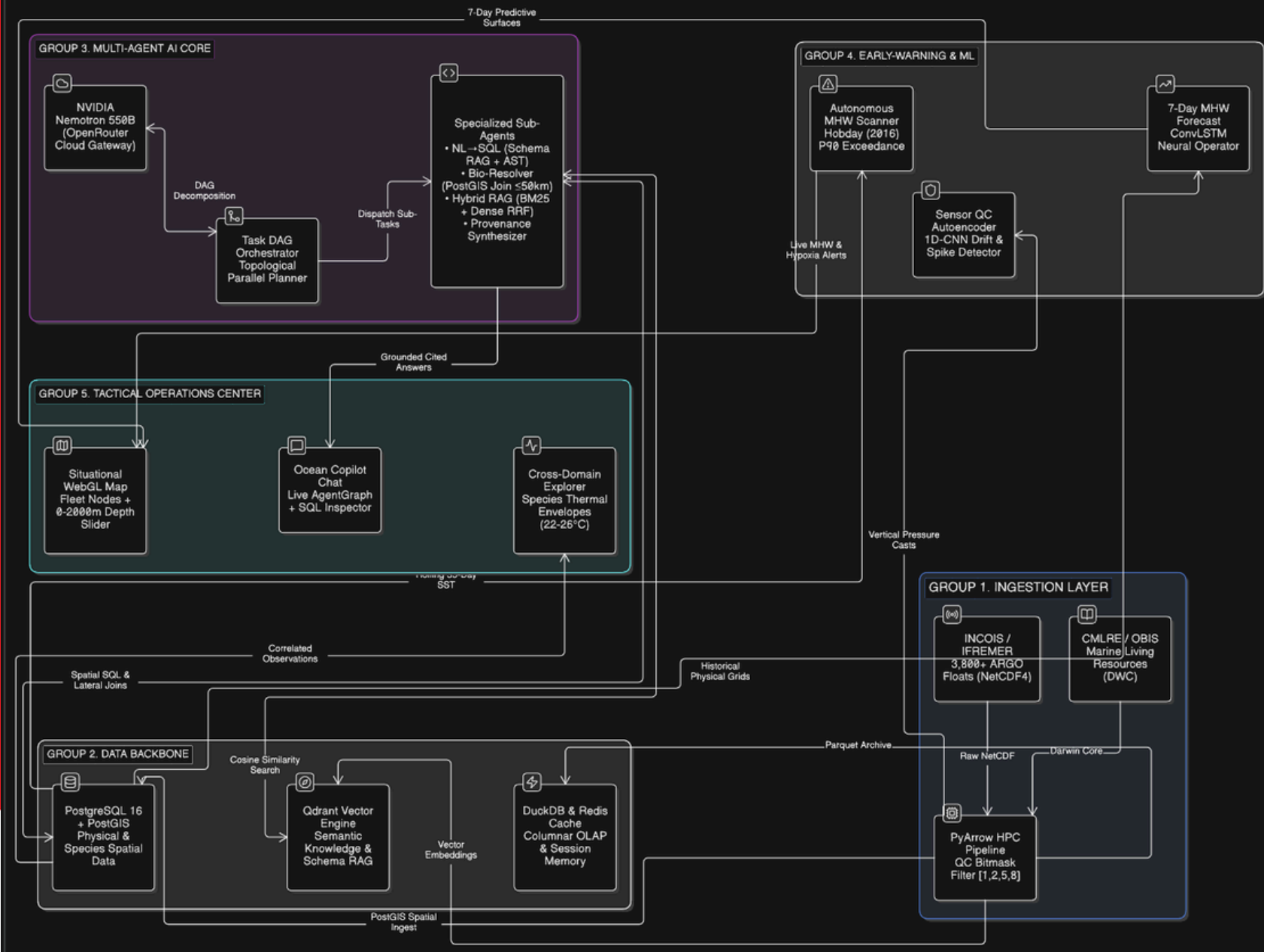
4. Handles Compound Questions

A planner decomposes multi-part cross-domain queries into sub-tasks instead of one-shot lookups.



Tech stack / Technology

VARUNA: National Marine Data Backbone & Multi-Agent Intelligence



Innovation / Differentiation

System Differentiation & Paradigm Shift

A rigorous architectural comparison between incumbent reactive, single-domain workflows and VARUNA's cohesive agentic and predictive capabilities to show its Innovation

Current System Limitations

System Component	Limitation
 Single-Shot RAG Pipeline	Incapable of resolving compound, multi-hop oceanographic queries
 Reactive Data Delivery	Portals are static; charts require manual expert initiation
 Disconnected Data Silos	Limited to isolated databases with no cross-domain biological fusion
 Raw Data Vulnerabilities	Blindly ingests raw sensor data, allowing biofouled measurements to pollute decisions

VARUNA Differentiators

Feature	Description
 Cognitive Multi-Agent Task DAG	An asynchronous orchestrator parses complex queries into an optimized, cycle-validated topological task graph
 Proactive Anomaly Forecasting	Background Spatio-Temporal ConvLSTM networks forecast Marine Heatwaves (MHW) and sea surface temperature anomalies 7 days in advance
 Cross-Domain Biology Fusion	Unifies physical INCOIS data with biological CMLRE data via Darwin Core metadata standards and asynchronous Spatial-Temporal Bounding Joins
 Self-Auditing Sensor Quality Control	An unsupervised 1D-CNN Autoencoder evaluates reconstruction error curves (0–2000m) to detect sensor drift, biofouling, and pressure spikes before data storage

FEASIBILITY / IMPACT / ROADMAP

Honest Scoping & Execution Proof

What VARUNA already runs end-to-end for the PoC, what is deliberately deferred, and the concrete safeguards against its two biggest technical risks.

Built for the PoC

Agentic / Predictive / Working

Multi-agent Task DAG

Planner, SQL-Gen, hybrid retrieval, entity resolution, comparator, synthesizer — a parallelized DAG, not single-shot.

Proactive anomaly engine

MHW + hypoxia detection running continuously on streams, not on demand.

7-day forecasting + sensor QC

ConvLSTM MHW forecasting paired with an autoencoder for sensor QC anomalies.

Cross-domain taxonomy correlation

OBIS/GBIF as a labeled CMLRE stand-in; bounded 50km / ± 7 -day window.

Full command-center frontend

Map, Analysis, Alerts, and Biodiversity views — live, not mockups.

Roadmapped

Deliberately deferred, not built

eDNA module

Not started, flagged as a future data source, out of scope for this PoC.

Full CMLRE production integration

Schema is Darwin Core-compatible a data-access wait, not a rebuild.

Otolith morphometrics CV module

Computer-vision age/growth analysis deferred to post-PoC phase.

Risks & mitigations

Hallucination on numbers

→ forced through SQL/tool-call path; provenance tracer rejects unsupported claims.

False positive anomalies/forecasts

→ validated against historical baselines (Gulf of Mannar 2020, six-basin event Apr 2026).

Cross-domain join imprecision

→ bounded to 50km / ± 7 -day windows, never exact-match.

Datasets: ARGO (INCOIS/IFREMER), OBIS/GBIF — public, no access blocker. **Impact:** faster, decision-ready, predictive ocean intelligence for MoES's national marine data backbone goal.

References

1. Hobday, A. J., Alexander, L. V., Perkins, S. E., Smale, D. A., Straub, S. C., Oliver, E. C. J., Benthuyssen, J. A., Burrows, M. T., Donat, M. G., Feng, M., Holbrook, N. J., Moore, P. J., Scannell, H. A., Sen Gupta, A., & Wernberg, T. (2016). A hierarchical approach to defining marine heatwaves. *Progress in Oceanography*, 141, 227–238.
<https://doi.org/10.1016/j.pocean.2015.12.014>
2. Katsogiannis-Meimarakis, G., & Koutrika, G. (2023). A survey on deep learning approaches for text-to-SQL. *The VLDB Journal*, 32(4), 905–936. <https://doi.org/10.1007/s00778-022-00776-8>
3. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.
<https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>
4. Qin, B., Hui, B., Wang, L., Yang, M., Li, J., Li, B., Geng, R., Cao, R., Sun, J., Si, L., Huang, F., & Li, Y. (2022). A survey on text-to-SQL parsing: Concepts, methods, and future directions. *arXiv*. <https://arxiv.org/abs/2208.13629>
5. Wieczorek, J., Bloom, D., Guralnick, R., Blum, S., Döring, M., Giovanni, R., Robertson, T., & Vieglais, D. (2012). Darwin Core: An evolving community-developed biodiversity data standard. *PLoS ONE*, 7(1), e29715.
<https://doi.org/10.1371/journal.pone.0029715>
6. Wu, Q., Bansal, G., Zhang, J., Wu, Y., Zhang, S., Zhu, E., Li, B., Jiang, L., Zhang, X., & Wang, C. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework. *arXiv*. <https://arxiv.org/abs/2308.08155>
7. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations (ICLR 2023)*.
<https://arxiv.org/abs/2210.03629>
8. OceanIQ. (n.d.). OceanIQ--AI-Intelligent [Source code repository]. GitHub. <https://github.com/PaarthNo1/OceanIQ--AI-Intelligent>
9. Ocean Biodiversity Information System. (n.d.). OBIS. <https://obis.org/>
10. Global Biodiversity Information Facility. (n.d.). GBIF. <https://www.gbif.org/>
11. Biodiversity Information Standards (TDWG). (n.d.). Darwin Core. <https://dwc.tdwg.org/>
12. Indian National Centre for Ocean Information Services. (n.d.). Argo data. <https://services.incois.gov.in/argo/>
13. Open Government Data Platform India. (n.d.). Indian Ocean Argo data. Ministry of Earth Sciences.
<https://www.data.gov.in/resource/indian-ocean-argo-data>
14. Coriolis Operational Oceanography. (n.d.). Data access and download. Ifremer. <https://www.coriolis.eu.org/Data-Access/Data-Download>

Team composition

Member no	Skills
Member 1 (Lead)	AI systems architecture — multi-agent orchestrator, anomaly engine, synthesizer, API gateway
Member 2	AI/Backend/data infrastructure — NetCDF ingestion, PostGIS schema, biodiversity seeding
Member 3	Predictive ML — MHW forecasting model, sensor QC autoencoder
Member 4	Frontend full-stack — chat interface, live agent DAG visualizer, 3D globe
Member 5	Geospatial/visualization — situational map, anomaly alert feed UI
Member 6	Cross-domain explorer, chart suite, PPT deck & video